

Problem Statement Selection

Problem statement addressed: Cleveland Clinic — *Unlocking undruggable targets: quantum simulation of allosteric signal propagation* (Global Quantum + AI Challenge 2026, Cleveland-Clinic-Challenge-Statement-vF-1.pdf).

How we interpreted it

Rather than only running the requested algorithm on the provided structures, our Phase-1 work audited the challenge’s own premise and dataset. Our measurements show that the standard benchmark targets conflate **static cavity retrieval** with **genuine cryptic-pocket discovery**: roughly half of apo/holo pairs cannot express the required contrast at all.

Our proposal therefore addresses the challenge’s core objective by delivering a **validated, unconfounded benchmarking instrument** capable of certifying allosteric methods — quantum or classical — alongside the required predictions.

Targets

We address the complete mandatory set required by Challenge Statement §6: **KRAS G12C, BCR-ABL1, cardiac myosin, and c-Myc**.

Four structure choices deviate from Table 1. **Each is stated, with its reason and its evidence, in Section 1 of the Concept Proposal**, and each traces to the organisers’ clarification of 2026-08-26. They are not repeated here: a single authoritative statement is less likely to drift than two, and the Concept Proposal is the document a reviewer assesses.

Required outputs

Challenge Statement §5’s three deliverables are supplied as follows:

§ 5 output	Where
Connectivity Matrix - N×N quantum connectivity strength	Connectivity_Matrices.csv
Hit List - top five predicted allosteric residues per target	Solution_Outputs.pdf, and machine-readable in the same file set
Methodological Report - the quantum metric chosen, and why it proxies biological signal transmission	Section 2 of the Concept Proposal, with per-target detail in Solution_Outputs.pdf